

How a Print Shop Works

The Complete Beginner-to-Advanced Study Guide

From zero knowledge to confident understanding —
written for software builders working with the print industry

- ✓ Every concept explained in simple English

- ✓ Real-world examples and analogies throughout

- ✓ The full job lifecycle: Order → Prepress → Press → Finishing → Delivery

- ✓ The business & software side: Estimating, Job Tickets, MIS, Web-to-Print

- ✓ Common mistakes, best practices, FAQs and a final cheat sheet

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How to use this guide: Read Parts 1–3 to understand how printing physically works. Part 4 is written especially for you as a solution builder — it explains the data, money and software flow. Part 5 is your quick-revision toolkit. Technical terms are always defined in simple language the first time they appear.

PART 1

FOUNDATIONS

What a print shop is, who works there, and how a job flows from order to delivery.

Chapter 1 — What Is Commercial Printing? The Big Picture

Simple definition: Commercial printing is the process of transferring digital files (designs, text, images) onto physical materials — paper, cardboard, plastic, fabric, vinyl — using professional presses and equipment, at scale. Unlike your home printer that makes 10 copies, a commercial print shop might produce 10,000 brochures, 500 banners, or 1 million labels.

Analogy: A Print Shop Is a Factory

Think of a print shop exactly like a small factory. Raw materials (paper, ink) come in one door. They pass through machines and skilled workers in a fixed sequence. A finished product (a book, a box, a business card) goes out the other door. Every concept in this guide — estimating, job tickets, scheduling, finishing — exists in some form in ANY factory. Print just has its own special vocabulary and machines.

What print shops produce

- **Marketing materials:** flyers, brochures, catalogs, posters, business cards, postcards
- **Publications:** books, magazines, newspapers, manuals, annual reports
- **Packaging:** boxes, cartons, labels, flexible pouches, food wrappers
- **Signage & large format:** banners, vehicle wraps, shop signs, exhibition displays
- **Stationery & forms:** letterheads, envelopes, invoices, certificates
- **Apparel & promo:** printed t-shirts, mugs, bags (screen printing / sublimation shops)

Types of print businesses

Type of shop	What they do	Typical equipment
Commercial printer	General marketing & publication work for businesses	Offset + digital presses, cutters, folders, binders
Digital / quick print	Short runs, fast turnaround, walk-in customers	Digital toner/inkjet presses, small finishers
Wide/large format shop	Banners, signs, wraps, displays	Roll-to-roll & flatbed inkjet printers, laminators, plotters
Label & packaging printer	Labels, cartons, flexible packaging	Flexo presses, die-cutters, folder-glueers
Screen printing shop	T-shirts, apparel, promotional items	Screen presses, dryers, heat presses
In-plant print center	Print department inside a company/university	Digital presses, internal billing systems
Print broker	Sells print but outsources production to trade printers	Software only — no presses!

Why this matters for your software: each shop type has different data needs. A label printer thinks in 'rolls' and 'dies'; a wide-format shop prices per square meter/foot; a commercial printer prices per sheet and per thousand. Good print software must model these differences.

Key Takeaways — Chapter 1

1) A print shop is a factory that turns digital files into physical products. 2) There are many shop types (commercial, digital, wide-format, label, screen, in-plant, broker) with different workflows. 3) Software for print must understand the shop type because pricing units and processes differ.

Chapter 2 — Inside a Print Shop: Departments, People and Machines

Walk into a mid-size print shop and you will find these departments, usually in this physical order from the front door to the loading dock:

1. Front office / Customer Service (CSR)

Customer Service Representatives (CSRs) take orders, answer 'where is my job?' calls, collect artwork files, and create the **job ticket** (the instruction sheet that travels with the job). The **estimator** also sits here — the person who calculates what a job will cost and what price to quote. In small shops, the owner does all of this.

2. Prepress department

Prepress operators check and fix customer files (**preflight**), arrange pages onto big press sheets (**imposition**), manage color, make **proofs** for customer approval, and output **plates** for offset presses or send files to digital press queues. This is the most computer-heavy department — and the one your software will talk to most.

3. Pressroom

Press operators run the printing machines. They mount plates, load paper, match colors against the approved proof, and watch quality throughout the run. A skilled offset press operator is one of the highest-paid roles in the shop.

4. Bindery / Finishing department

After printing, sheets go here to be cut, folded, stitched, glued, laminated, or die-cut into the final product. Finishing has the most manual labor and — in many shops — the most bottlenecks.

5. Shipping / Dispatch

Finished goods are counted, boxed, labeled and either delivered, couriered, or collected by the customer. Inventory of paper and supplies is also managed near here (the paper store).

Analogy: A Restaurant

Front office = waiters taking your order. Prepress = the kitchen prep station washing and chopping ingredients. Pressroom = the chefs cooking on the stove. Finishing = plating and garnishing the dish. Dispatch = the waiter delivering it to your table. If the prep station passes bad ingredients (bad files), the chefs cannot fix it — same in print: problems caught late are expensive.

Key roles you should know

Role	What they do	Software they touch
Estimator	Calculates cost & quote for each job	Estimating module of MIS
CSR (Customer Service Rep)	Order entry, customer communication, job tickets	CRM, order management
Prepress operator	Preflight, imposition, proofs, plates	Preflight tools, RIP, imposition software
Press operator	Runs the press, controls color & registration	Press console, job queue

Role	What they do	Software they touch
Bindery operator	Cutting, folding, binding, finishing	Job tickets, barcode scanners
Production manager	Schedules all jobs across machines & people	Scheduling board / MIS dashboard
Owner / GM	Pricing strategy, profitability, cash flow	Reports, invoicing, accounting

Key Takeaways — Chapter 2

1) Five core areas: Front office → Prepress → Pressroom → Finishing → Dispatch. 2) The job ticket is the instruction sheet that travels with every job. 3) Each role interacts with different software modules — map your solution's features to these real roles.

Chapter 3 — The Life of a Print Job (The Master Workflow)

This is the single most important chapter in the guide. Every print job in every shop follows the same basic path: an order comes in, files get checked, the job is set up, printed, finished, and shipped. Memorize this flow — every feature of print software exists to support one of these steps.

1. Enquiry & Estimate

Customer asks 'how much for 5,000 brochures?' Estimator calculates material + machine + labor cost, adds margin, sends a quote.

2. Order & Job Ticket

Customer accepts. The quote converts into a JOB with a unique job number and a job ticket listing every specification.

3. Artwork & Preflight

Customer uploads files. Prepress runs preflight checks: resolution, colors, fonts, bleed. Problems are fixed or sent back.

4. Proof & Approval

A proof (PDF or printed sample) goes to the customer. Nothing prints until the customer approves. This protects the shop legally.

5. Imposition & Plates/RIP

Pages are arranged onto the big press sheet layout. For offset: plates are made. For digital: the file is RIPped to the press queue.

6. Printing (Press)

The press is set up ('make-ready'), colors are matched to the proof, and the run is printed with quality checks throughout.

7. Finishing (Postpress)

Sheets are cut, folded, bound, laminated, die-cut — whatever the job ticket specifies — becoming the final product.

8. QC, Packing & Dispatch

Final quality check, counting, boxing, labeling, delivery. Job is marked complete.

9. Invoice & Costing

Invoice is sent. Actual costs (time, materials used) are compared to the estimate to see if the job made money.

Two critical ideas hidden in this flow

1) Handoffs are where jobs die. Between every step, someone must move files, re-enter data, or chase an approval. In manual shops each handoff means email, paper, and human memory — the source of most errors and delays. Print workflow automation exists to replace these manual handoffs with software.

2) Errors get 10× more expensive at every stage. A wrong color spotted at preflight costs 5 minutes. Spotted on the proof: an email. Spotted on press: wasted plates and paper. Spotted after delivery: a full reprint plus an angry customer. This is why prepress is treated as quality control, not clerical work.

Real-World Example

A café orders 2,000 tri-fold menus. (1) Estimator quotes ■18,000 based on art paper 170 GSM, 4-color both sides, folding. (2) Café accepts → Job #4512 created. (3) Preflight finds the logo is RGB and a photo is 72 DPI → asks designer for fixes. (4) PDF proof approved on Tuesday. (5) 8 menus are imposed per SRA3 sheet → 250 sheets needed + waste. (6) Digital press prints 275 sheets (25 extra for setup/waste). (7) Cut to size, creased, folded. (8) Boxed and delivered Thursday. (9) Invoice sent; actual cost ■13,200 → profit visible in the system.

Key Takeaways — Chapter 3

1) Every job: Estimate → Order → Preflight → Proof → Imposition → Press → Finishing → Dispatch → Invoice. 2) Manual handoffs between steps cause most errors — automation targets handoffs. 3) The earlier a problem is caught, the cheaper it is to fix (the 10× rule).

PART 2

THE LANGUAGE OF PRINT

Color, files, and paper — the vocabulary every print job specification is written in.

Chapter 4 — Color: CMYK, RGB, Spot Colors and Pantone

RGB — the color of screens

Screens (phones, monitors) create color with light: **R**ed, **G**reen and **B**lue. Adding all three at full strength makes white. This is called an **additive** color system. RGB can show very bright, glowing colors — because light glows.

CMYK — the color of ink

Presses create color with four inks: **C**yan (blue), **M**agenta (pink-red), **Y**ellow, and **K**ey (black). Inks work by absorbing light, so this is a **subtractive** system. Tiny dots of the four inks printed in overlapping patterns trick your eye into seeing millions of colors. 'K' is used for black because it is the 'Key' plate that other colors align to (and to avoid confusion with B for blue).

Analogy: Flashlights vs Paint

RGB is like mixing colored flashlights on a wall — more light = brighter, all together = white. CMYK is like mixing paint — more paint = darker and muddier. This is why a glowing neon blue on screen turns dull navy when printed: paint physically cannot glow. Designers must convert RGB → CMYK and check colors before printing, or use a spot color for that effect.

Spot colors and Pantone (PMS)

Instead of building a color from 4 inks, a **spot color** is one pre-mixed ink of exactly the right shade — like buying a ready-made paint color instead of mixing it yourself. The **Pantone Matching System (PMS)** is the world-standard catalog of these colors (e.g., 'Pantone 485 C' is a specific red). Brands use Pantone so their logo red is identical in every country and every print shop. Offset presses handle spot colors well; many digital presses can only simulate them with CMYK.

Color terms you will hear

Term	Simple meaning
Gamut	The range of colors a device can reproduce. RGB gamut is wider than CMYK — some screen colors are simply unprintable.
Color management / ICC profiles	Software 'translation dictionaries' that keep color consistent from screen → proof → press.
4/4, 4/0, 1/0	Shorthand for colors per side. 4/4 = full color both sides; 4/0 = full color front, blank back; 1/0 = one color front only.
Rich black	Black made of all four inks (e.g., 60C 40M 40Y 100K) for deep solid areas. Plain 100K black can look washed out on large areas.
Registration	The precise alignment of the 4 ink layers. 'Out of register' = blurry, rainbow-edged print.
Overprint	Printing one ink on top of another instead of 'knocking out' the shape beneath — black text is usually set to overprint.
Press check	Customer/designer stands at the press when the run starts to approve color on the actual printed sheets.

Key Takeaways — Chapter 4

1) RGB = light/screens, CMYK = ink/print; files must be CMYK for press. 2) Vivid RGB colors can dull when converted — check or use Pantone spot inks. 3) 4/4, 4/0 shorthand describes colors per side and directly drives price.

Chapter 5 — Files & Layout: Bleed, Trim, Crop Marks, DPI, Vector vs Raster

Why files matter so much

The #1 source of delays and disputes in any print shop is bad customer files. Understanding these few concepts explains almost every preflight error your software will ever need to display.

Trim, bleed and safe zone

Printers do not print on the final-size paper. They print on **larger sheets** and then cut ('trim') down to size. Because cutting machines can be off by a millimeter, three invisible lines govern every design:

- **Trim line:** where the cut is supposed to happen — the final size of the piece.
- **Bleed:** background colors/images must extend ~3 mm (1/8 inch) **BEYOND** the trim line. If the cut drifts slightly, there is still ink there — no ugly white sliver at the edge.
- **Safe zone (margin):** keep text and logos ~3–5 mm **INSIDE** the trim line, so nothing important gets chopped off.
- **Crop marks:** small lines printed at the corners showing the cutter operator exactly where to cut. They are trimmed away.

Analogy: Cutting a Cake

Imagine icing a cake that will be cut to a smaller square. If you ice only up to the exact cut line and your knife drifts 1 mm outward, guests see bare cake (white paper). So you ice a little **PAST** the line — that extra icing is the bleed. And you don't write 'Happy Birthday' right on the cut line — you keep it safely inside. That's the safe zone.

Resolution: DPI and PPI

DPI (dots per inch) measures how much detail an image has when printed. The print standard is **300 DPI at final size**. Web images are usually 72 DPI — they look fine on screen but print blurry and 'pixelated' (blocky). This is the single most common preflight failure. Note: resolution is at **FINAL SIZE** — a 300 DPI image enlarged 4× becomes 75 DPI.

Vector vs Raster

- **Raster** images (JPEG, PNG, photos) are grids of pixels. Enlarging them loses quality.
- **Vector** graphics (logos made in Illustrator; AI, EPS, SVG, PDF shapes) are mathematical curves. They scale to any size — business card or building banner — with zero quality loss. Logos and text should always be vector.

The print-ready PDF

The universal delivery format for print is PDF, ideally the **PDF/X** standard (PDF/X-1a is strictest: all CMYK, all fonts embedded; PDF/X-4 allows modern transparency). 'Fonts embedded' means the font data travels inside the file — otherwise the print shop's computer substitutes a different font and the layout breaks. A print-ready PDF has: correct size, 3 mm bleed, crop marks, CMYK colors, 300 DPI images, embedded fonts.

Key Takeaways — Chapter 5

1) Bleed outside the trim, text inside the safe zone, crop marks show where to cut. 2) 300 DPI at final size for print; 72 DPI web images fail. 3) Logos = vector; delivery format = print-ready PDF (PDF/X) with fonts embedded.

Chapter 6 — Paper & Substrates: GSM, Stocks, Coatings, Grain

Substrate is the industry word for 'the material you print on' — paper, board, vinyl, plastic film, fabric, metal. For most shops it is paper, and paper is usually the single largest cost in a job. Three choices define any paper: **stock (type), weight, and finish**.

Weight: GSM (and lb / pt)

GSM = grams per square meter — the weight of a 1m × 1m sheet. Higher GSM = thicker, stiffer paper. It is the universal, unambiguous measure. The US also uses pounds (lb) — confusingly, 80 lb text and 80 lb cover are totally different because they reference different base sheet sizes; and 'points' (pt) measure physical thickness (1 pt = 1/1000 inch). When in doubt, use GSM.

GSM range	Feels like	Typical use
70–100 GSM	Copier / letterhead paper	Book inner pages, letterheads, forms
115–170 GSM	Flexible but substantial	Flyers, posters, magazine pages, brochures
200–300 GSM	Card, holds its shape	Covers, postcards, folders, greeting cards
300–400 GSM	Rigid board	Business cards, packaging, premium covers

Stock types and finishes

- **Coated art paper (gloss/silk/matte):** smooth clay-coated surface; ink sits on top → sharp, vibrant photos. Gloss shines, matte doesn't glare.
- **Uncoated:** natural, absorbent surface (like copier paper); softer look, easy to write on, colors slightly muted.
- **Specialty:** textured, linen, kraft (brown craft look), recycled, synthetic (waterproof), self-adhesive label stock.

Grain direction

Paper fibers align in one direction during manufacturing. Folding **WITH** the grain is smooth; folding **AGAINST** it cracks — especially on heavy coated stocks (which is why thick covers are **scored/creased** before folding). Imposition software considers grain when laying out jobs.

How paper drives cost (crucial for estimating)

Paper is bought in large 'parent sheets' or rolls. The estimator calculates how many finished pieces fit on each press sheet ('N-up'), how many sheets the run needs, plus **waste/overs** (extra sheets for machine setup, typically 2–10%). A clever imposition that fits 9-up instead of 8-up on the sheet cuts paper cost by ~11% — this is why estimating software and imposition are tightly linked.

Key Takeaways — Chapter 6

1) Substrate = the material printed on; paper is the biggest material cost. 2) GSM is the universal weight measure (300 GSM business card vs 90 GSM letterhead). 3) Coated = vibrant photos; uncoated = natural/writable. Heavy stock must be creased before folding. 4) Sheet layout efficiency (N-up) and waste allowance drive the paper calculation in every estimate.

PART 3

THE THREE STAGES OF PRODUCTION

Prepress → Press → Postpress. How files become plates, plates become prints, and prints become products.

Chapter 7 — Prepress: Preflight, Proofing, Imposition and RIP

Prepress is everything between 'customer approved the design' and 'ink hits paper'. It is where most quality problems are caught — cheaply. A modern prepress workflow has five steps:

Step 1 — Preflight (file checking)

Named after a pilot's pre-flight checklist. Automated software (e.g., built into Enfocus PitStop, Adobe Acrobat, or workflow systems) inspects the file for:

- Images below 300 DPI (the most common finding)
- RGB or wrong colors that should be CMYK / spot
- Missing or non-embedded fonts
- Missing bleed, wrong page size or page count
- Hairlines too thin to print, overprint mistakes, transparency issues

Problems are auto-fixed where possible or reported back to the customer. **For your software:** preflight status (pass / fixed / rejected) is a key job state to track and communicate.

Step 2 — Color management

Using ICC profiles and calibrated devices so that the screen, the proof, and the press all produce matching color. Without it, the customer approves one blue and receives another.

Step 3 — Proofing & approval

- **Soft proof:** a PDF viewed on screen — fast, free, fine for most work.
- **Hard proof / contract proof:** a physical, color-accurate print the customer signs. 'Contract' because it becomes the legal color standard the press run must match.
- Online approval portals (part of web-to-print / MIS) timestamp the approval — vital for dispute protection.

Step 4 — Imposition

Arranging multiple pages or copies onto one large press sheet in the correct position and rotation, so that after printing, folding and cutting, everything ends up in the right order. A 16-page booklet is not printed page by page — 8 pages are imposed on each side of a big sheet, which is folded into a **signature**. Imposition also does **ganging**: combining different small jobs (e.g., 6 different business cards) onto one sheet to share the cost.

Analogy: Origami Planning

Imposition is like planning an origami. You must print page 16 upside-down next to page 1, because after three folds of the big sheet, they land in perfect reading order. Software does this math from rules: sheet size, press, folding scheme, binding style and grain direction.

Step 5 — RIP and plate-making

The **RIP (Raster Image Processor)** converts the smooth PDF into the actual grid of ink dots the press will fire — including **halftone screening**: simulating shades using dot sizes (look at a newspaper photo with a magnifier — it's all dots). For offset, a **CTP (Computer-to-Plate)** machine then laser-etches each color's image onto an aluminum plate. For digital presses, the RIPped file goes straight into the press queue —

no plates.

Key Takeaways — Chapter 7

1) Prepress = preflight → color management → proof/approval → imposition → RIP/plates. 2) Preflight catches the classic errors (low DPI, RGB, missing fonts, no bleed). 3) Nothing prints before approval; the contract proof is the color standard. 4) Imposition (N-up, signatures, ganging) turns pages into efficient press-sheet layouts.

Chapter 8 — The Press: Offset, Digital, Flexo, Screen, Gravure & Wide-Format

Different machines put ink on substrate in fundamentally different ways. The choice depends on quantity, quality, material and whether every copy is identical. This decision drives the entire cost structure of a job.

Offset lithography — the workhorse of commercial print

How it works: the image is laser-etched onto an aluminum **plate** (one per ink color). On press, the plate is dampened with water and inked — ink (oily) sticks only to image areas, water repels it elsewhere. The inked image transfers ('offsets') first onto a rubber **blanket** cylinder, then from the blanket onto the paper. Sheet-fed offset presses print cut sheets; web offset presses print from huge continuous rolls (newspapers, magazines).

- **Strengths:** superb quality, exact Pantone colors, lowest cost per copy on long runs (1,000s+).
- **Weaknesses:** expensive setup ('make-ready': plates, wash-up, waste sheets) — uneconomical for 50 copies; every copy identical.

Digital printing — no plates, instant start

The file goes straight from the RIP to the press. Two families: **toner** (like a giant laser printer — electrostatic drum + powdered toner, e.g., Xerox/Ricoh/Canon production presses) and **inkjet** (microscopic ink droplets fired from printheads, increasingly used for high-speed book and mail production).

- **Strengths:** zero setup → cheap short runs; first copy = final quality; fast turnaround; **Variable Data Printing (VDP)** — every copy can be different (names on certificates, personalized mail, numbered tickets).
- **Weaknesses:** higher cost per copy at volume (the 'click charge' per page), slightly narrower color/materials range.

The Break-Even Rule (memorize this)

Digital and offset costs cross around 500–2,000 copies depending on the job. Below the break-even → digital wins (no setup cost). Above it → offset wins (setup cost is spread across thousands of copies, and each copy is cheaper). Every estimating engine in print software is essentially finding this break-even for each job.

Flexography (flexo) — the packaging champion

Uses flexible rubber/photopolymer **relief plates** (raised image, like a stamp) wrapped on fast-spinning cylinders. An engraved **anilox roller** meters ink onto the plate, which stamps it directly onto the substrate — which can be almost anything flexible: plastic film, foil, corrugated board, label rolls. Very high speed, roll-to-roll, cheap fast-drying inks. Dominates labels, food wrappers, boxes and bags. Quality is slightly below offset for fine detail.

Screen printing

Ink is squeezed through a fine mesh **stencil (screen)** — one screen per color — onto t-shirts, posters, signage, glass, metal. Deposits very thick, opaque, durable ink layers (that's why printed t-shirt graphics feel raised). Manual or automatic carousel presses; slow but unbeatable on fabric and odd surfaces.

Gravure and others

Gravure: image engraved as tiny cells INTO a copper cylinder; excellent consistency for multi-million runs (magazines, flexible packaging) but cylinders are very expensive. **Letterpress:** the historic raised-type method, now a premium craft look (wedding cards). **Sublimation/DTG/DTF:** modern garment-decoration methods.

Wide / large format inkjet

Roll-to-roll printers (up to several meters wide) print banners, vinyl, wallpapers; **flatbed** printers print directly onto rigid boards (acrylic, foam board, wood) using UV-cured inks. Priced per square meter/foot. Finishing here means laminating, contour-cutting on a plotter, welding hems and adding eyelets — different world from paper bindery.

Choosing the method — summary table

Method	Best for	Run length	Key limitation
Offset (litho)	Brochures, books, magazines, packaging board	1,000 – millions	Setup cost kills short runs
Digital toner/inkjet	Short runs, personalization, fast jobs	1 – ~2,000	Per-copy 'click' cost at volume
Flexo	Labels, flexible packaging, corrugated	10,000 – millions	Fine detail below offset
Screen	Apparel, promo items, special surfaces	10 – 1,000s	Slow; one screen per color
Gravure	Very long packaging/magazine runs	Millions	Very expensive cylinders
Wide-format inkjet	Banners, signs, displays, wraps	1 – 100s	Priced per area; slow per unit

Key Takeaways — Chapter 8

1) Offset: plate → blanket → paper; best quality & cost at volume. 2) Digital: no plates; wins short runs and enables variable data (VDP). 3) Flexo rules packaging/labels; screen rules apparel; wide-format rules signage. 4) The digital-vs-offset break-even (~500–2,000 copies) is the core logic of print estimating.

Chapter 9 — Postpress & Finishing: Cutting, Folding, Binding, Coatings

Finishing ('bindery', 'postpress', or 'converting' in packaging) transforms printed sheets into finished products. It can happen **in-line** (units attached to the press) or **off-line** (separate machines). Many jobs pass through 3–5 finishing operations — each one a line on the job ticket and a cost in the estimate.

Cutting family

- **Guillotine cutting/trimming:** a heavy blade slices stacks of sheets at marked positions — the most universal operation.
- **Die-cutting:** a custom-shaped steel blade ('die' — like a cookie cutter) punches shapes: folder pockets, box blanks, door hangers, round labels. The **dieline** is the template drawing of all cuts and creases.
- **Contour/kiss cutting:** plotter-cut shapes in wide format; kiss cutting cuts the sticker but not its backing paper.
- **Perforating:** dotted line of tiny holes for tear-off coupons and tickets. **Drilling/punching:** holes for binders and coils.

Folding family

- **Folding:** half-fold, tri-fold (letter fold), Z-fold, gate fold — machine folders do thousands per hour.
- **Scoring/creasing:** pressing a hinge line into thick or coated stock BEFORE folding so it doesn't crack — mandatory on digital prints (toner cracks) and heavy covers.

Binding family (fastening pages)

Method	How it works	Best for
Saddle stitching	Folded sheets stapled through the spine fold (2 wire staples)	Booklets, magazines up to ~64 pages; cheap & fast
Perfect binding	Pages glued into a wrap-around cover with a flat, printable spine	Paperback books, catalogs, thicker documents
PUR binding	Perfect binding with stronger polyurethane glue	Heavy-use catalogs, lay-flatter, coated stocks
Spiral / Wire-O	Holes punched; plastic coil or metal wire threaded through	Manuals, workbooks, calendars — opens 360°, lies flat
Section sewn (Smyth)	Signatures sewn with thread, then glued/cased	Hardcovers, premium books — most durable, lies flat
Case binding	Sewn or glued block attached into rigid hardcover boards	Hardcover books

Surface finishing (protection + wow factor)

- **Lamination:** a plastic film heat-bonded over the sheet — gloss, matte, or velvety **soft-touch**. Maximum protection.
- **UV coating:** liquid coating cured instantly by UV light; **spot UV** puts glossy highlights only on chosen areas (logo pop).
- **Aqueous/varnish:** lighter, cheaper protective coatings applied on-press.

- **Foil stamping:** metallic foil pressed on with a heated die — gold/silver luxury look.
- **Embossing/debossing:** raising or sinking the paper surface for a tactile 3D effect.

Assembly & converting

Collating (gathering pages in order), **folder-gluing** (folding and gluing die-cut box blanks into cartons), **padding** (glue-edge notepads), **shrink-wrapping**, kitting and mailing operations (addressing, inserting, franking).

Why finishing matters to your software

Finishing is where scheduling breaks down: it's the most manual, most sequential stage — a job might need cut → crease → fold → stitch → trim on 4 different machines with 4 operators. Each operation needs: estimated time, machine/operator assignment, status capture (often barcode scans), and its own cost rate. Model finishing as a LIST of operations per job, not a single step.

Key Takeaways — Chapter 9

1) Finishing = cutting, folding/creasing, binding, coating, converting — usually several per job. 2) Saddle stitch (thin booklets), perfect bind (books), coil (lay-flat manuals), sewn (premium). 3) Crease before folding heavy/coated/digital stock or it cracks. 4) Lamination/UV/foil/emboss add protection and premium feel — and revenue.

PART 4

THE BUSINESS & SOFTWARE SIDE

How print jobs are priced, tracked and automated — the domain knowledge behind every Print MIS.

Chapter 10 — Estimating & Quoting: How a Print Job Is Priced

Estimating is the heart of a print business — and of print software. Printing is custom manufacturing: almost every job is different, so there is no fixed price list for most work. The estimator (or the estimating engine) rebuilds the cost of every job from its specifications.

The specification (the inputs)

A complete job spec answers: **What product? Final size? Page count? Quantity? Paper stock & GSM? Colors (4/4? spot?)? Printing method? Finishing operations? Packing & delivery? Deadline?** Change any one and the price changes.

The cost build-up (the calculation)

Cost block	What goes in it	How it's calculated
Materials	Paper/substrate, ink/toner clicks, plates, lamination film, boxes	Sheets needed = quantity ÷ N-up + waste allowance; × cost per sheet
Prepress	Preflight/artwork time, proofs, plate output	Time × hourly rate + plate/proof unit costs
Press	Make-ready (setup) + running time	Setup time × press hourly rate + (sheets ÷ speed) × rate; digital = clicks × click rate
Finishing	Each operation: cutting, folding, binding...	Per operation: setup + (units ÷ speed) × machine/labor rate
Outsourcing	Work sent to trade suppliers (e.g., foiling)	Supplier quote + handling margin
Delivery & overhead	Packing, transport, admin share	Flat fees or percentages

Then: **Cost + Margin = Price**. Margin may vary by customer, product or urgency. Good systems also show **price breaks** (500 / 1,000 / 2,500 quantities) because unit price falls as quantity rises — setup costs get shared.

Worked Mini-Example: 1,000 A5 flyers, 4/4, 170 GSM gloss, digital

A5 flyers impose 8-up on SRA3 sheets → $1,000 \div 8 = 125$ sheets + 5% waste ≈ 132 sheets. Paper: $132 \times \text{■}9 = \text{■}1,188$. Clicks: $132 \text{ sheets} \times 2 \text{ sides} \times \text{■}18 = \text{■}4,752$. Cutting: $15 \text{ min} \times \text{■}600/\text{hr} = \text{■}150$. Prepress/proof: $\text{■}300$. Total cost ≈ $\text{■}6,390$. Margin 40% → quote ≈ $\text{■}8,950$. Notice how N-up, waste %, click rate and hourly rates are all configurable numbers — these are exactly the 'rate cards' a Print MIS stores.

Why estimating is hard (and valuable)

- It must compare routes: digital vs offset, 8-up vs 9-up, in-house vs outsource — and pick the cheapest valid one.
- Rates go stale: paper prices change monthly; wrong rates silently destroy margins.
- Consistency: two human estimators often price the same job differently; software makes pricing consistent and instant.
- Estimate vs Actual: after the job, comparing estimated cost to actual recorded cost reveals real profitability and bad rates.

Key Takeaways — Chapter 10

1) $\text{Price} = \text{Materials} + \text{Prepress} + \text{Press} + \text{Finishing} + \text{Outsource} + \text{Delivery} + \text{Margin}$. 2) The paper calc (N-up + waste) and the setup-vs-running split are the core formulas. 3) Estimating engines are rate-card driven — configurable machine speeds, hourly rates, click charges, waste %. 4) Estimate-vs-actual comparison closes the loop on profitability.

Chapter 11 — Job Tickets, Scheduling and Shop-Floor Tracking

The job ticket

When a quote is accepted, it becomes a **job** with a unique job number, and the system generates the **job ticket** (also called work order or docket): the complete instruction sheet for production. Traditionally a paper envelope physically traveling with the job; in modern shops, a digital ticket on screens/tablets. It contains: customer & delivery info, specs (size, stock, colors), the operation list in sequence, quantities at each stage, proof status, special instructions, and the deadline.

Scheduling

The production manager must fit dozens of jobs onto limited machines and people, honoring deadlines. Good scheduling:

- Sequences jobs on each machine (press, cutter, folder, stitcher) by due date, priority and capacity.
- **Batches similar jobs** (same stock, same size) together to minimize changeover/setup time.
- Detects conflicts and bottlenecks BEFORE they happen (two jobs on one press at the same time; finishing overload on Thursday).
- Recalculates when reality changes — a press breakdown or a rush job reshuffles everything.

Shop-floor data collection (tracking)

To know where every job is, shops capture status at each workstation — classically by **barcode scanning** the job ticket at start/finish of each operation (or tapping on a tablet/phone app). This yields:

- **Real-time job status** — no more walking the floor or 'where is my order?' calls; customers can even self-track online.
- **Actual times and materials** per operation → feeds estimate-vs-actual costing.
- **Operator/machine performance data** → find bottlenecks, justify new equipment.
- **Quality & traceability records** — which paper batch, which operator, which machine (vital in packaging).

Software design insight

A print job is best modeled as: JOB → ordered list of OPERATIONS → each with machine, operator, estimated time, actual time, status, and material consumption. The schedule is a constraint problem over machines-as-resources. Status events (scan in/out) are your telemetry stream. Nearly every MIS is built on this skeleton.

Key Takeaways — Chapter 11

1) The job ticket is the single source of truth traveling with each job. 2) Scheduling = sequencing jobs on machines, batching similar work, killing bottlenecks early. 3) Barcode/app scans at each station give live status + actual costs. 4) Model: Job → Operations → Events. This is the data backbone of print software.

Chapter 12 — Print MIS, ERP, Web-to-Print and Automation (JDF, PDF/X)

The software map of a print shop

System	What it does	Examples
Print MIS	Management Information System — the business backbone: CRM, estimating, quoting, job tickets, scheduling, inventory, purchasing, dispatch, invoicing, reports	PrintIQ, Pace/EFI, DocketManager, Ordant, PrintPLANR, Printavo (screen shops)
Print ERP	MIS + full accounting, HR, supply chain, BI — for larger enterprises	EFI/ePS suites, Avanti
Web-to-Print (W2P)	Online storefronts (B2C shop or B2B corporate portals) where customers order, upload art, personalize templates and pay — orders flow straight into the MIS	OnPrintShop, PrintNow, Gelato, Vpress
Prepress workflow	Preflight, color, imposition, RIP, press queues	Kodak Prinergy, Heidelberg Prinect, Esko, Fiery, PitStop
Design/editor tools	Online template editors for personalization	Chili Publish, canvas-based editors
Shipping/accounting integrations	Rates, labels, ledgers	Carrier APIs, QuickBooks/Xero/Tally

How they connect: the automation dream

The ideal flow: a customer orders on the W2P store → the MIS creates the job and schedules it → files auto-preflight → auto-imposition gangs it with similar jobs → the RIP queues it to a press → barcode scans track finishing → shipping label prints → invoice raises itself. Human touches only exceptions. Shops call this 'touchless' or 'lights-out' workflow; even partial automation (say, auto-preflight + online approval) removes big bottlenecks.

The data standards that glue it together

- **PDF/X:** the standardized print-ready file format (X-1a strict CMYK; X-4 modern with transparency).
- **JDF (Job Definition Format):** an XML standard describing a print job's specs and process steps so MIS, prepress and presses from different vendors can exchange job data — 'the job ticket as XML'.
- **JMF (Job Messaging Format):** JDF's companion for live messages — machine status, counts, completion signals back to the MIS.
- APIs/webhooks are increasingly replacing raw JDF in cloud systems, but JDF remains widespread on press hardware.

Why shops buy (and what wins deals)

- Without MIS: quoting from spreadsheets, whiteboard scheduling, paper tickets, re-typing data 5 times → typical shops lose 5–20% of revenue to inefficiency.
- With MIS: consistent instant quotes, real-time visibility, fewer errors/reprints, faster throughput, known job profitability.

- Buyers care about: estimating accuracy for THEIR processes, ease of use for non-technical staff, integrations (prepress, accounting, carriers), and cloud access from anywhere.

Key Takeaways — Chapter 12

1) MIS = business backbone; W2P = online order intake; prepress workflow = file automation; they must integrate. 2) JDF/JMF (XML) and PDF/X are the interchange standards; APIs are the modern glue. 3) The value story: kill manual handoffs → fewer errors, faster jobs, visible profit. 4) Your solution should slot into this map — know which boxes you replace and which you integrate with.

PART 5

MASTERY

Mistakes to avoid, habits of great shops, answers to common questions — and your one-page revision sheet.

Chapter 13 — Common Mistakes and How to Avoid Them

File & prepress mistakes (the classics)

Mistake	Consequence	Prevention
RGB files sent to press	Colors shift dull/wrong	Preflight auto-detects; convert to CMYK; educate customers
72 DPI web images	Blurry, pixelated print	Enforce 300 DPI at final size in preflight
No bleed	White slivers at trimmed edges	Require 3 mm bleed; reject or fix at preflight
Fonts not embedded	Font substitution, broken layout	Demand PDF/X; embed or outline fonts
Text too close to edge	Words chopped off in trimming	Enforce the safe zone (3–5 mm)
Skipping the proof	Customer disputes the result; shop pays for reprint	Never print without recorded approval

Production mistakes

- **Printing before approval** or from an old file version — version control and locked approvals prevent this.
- **No creasing on heavy/coated/digital stock** → cracked folds on 5,000 finished brochures.
- **Ignoring grain direction** → rough folds, curling covers, jamming sheets.
- **No waste allowance** in the paper order → run ends 80 sheets short; press stops; paper reordered at panic prices.
- **Color judged under wrong light** → proofs and press sheets are compared in standard D50 viewing booths for a reason.

Business mistakes

- **Stale rate cards:** quoting on last year's paper prices silently turns profitable jobs into losses.
- **Underquoting complexity:** forgetting a finishing operation in the quote — it still must be done, unpaid.
- **No estimate-vs-actual review:** the shop never learns which jobs or customers lose money.
- **Scheduling in someone's head:** when that person is sick, the shop is blind. Systems, not heroes.
- **Accepting any file 'to be nice':** fixing terrible files free of charge is invisible unpaid labor; charge for artwork time.

Key Takeaways — Chapter 13

The same ten errors cause most losses in most shops: RGB, low-res, no bleed, no embed, no proof, no crease, no waste allowance, stale rates, forgotten operations, invisible artwork labor. Every one is preventable by process + software.

Chapter 14 — Best Practices of Well-Run Print Shops

- **Standardize everything:** templates for common jobs, fixed file-intake specs, consistent box labeling, tools ready at each station. In a truly standardized shop, workers can swap stations with minimal disruption.
- **Draw the workflow:** literally flowchart every step of the shop's process to expose problem areas — the first step of any improvement (and of any software implementation).
- **Automate the handoffs first:** start with the biggest bottleneck (usually file intake + preflight + approval), not with everything at once.
- **Batch and gang:** group jobs by stock/size to slash setup time and paper waste.
- **Measure estimate vs actual on every job** and review weekly — this is how pricing gets sharp.
- **Single source of truth:** one system holding job status, visible to office AND floor AND customers; kill the parallel spreadsheets.
- **Maintain machines preventively:** a press down for a day scrambles every schedule; maintenance windows belong in the schedule.
- **Continuous improvement culture (the Toyota lesson):** small experiments, constant refinement — 'don't be afraid to fail, be afraid to become stale.'
- **Sustainability as practice and pitch:** FSC-certified paper chains, waste reduction, aqueous coatings — customers ask for it.

Key Takeaways — Chapter 14

Standardize → visualize → automate bottlenecks → batch work → measure actuals → one source of truth → maintain machines → improve continuously. Software enables all eight, but process discipline comes first.

Chapter 15 — Frequently Asked Questions

Q: Is offset printing dying because of digital?

A: No. Digital keeps taking short-run work, but offset still wins on cost and quality for long runs, and flexo/gravure dominate packaging. They coexist; many shops run both and route each job to the cheaper method.

Q: Why can't the shop just print my Word/Canva file?

A: Those tools are RGB, low-res and don't manage bleed/fonts reliably. Shops convert them to print-ready PDFs — which takes chargeable artwork time and can shift colors.

Q: Why does 500 copies cost almost the same as 1,000?

A: Setup (make-ready, plates, file work) is a fixed cost. Doubling the run only adds paper + run time, so unit price drops steeply with quantity.

Q: What's the difference between a proof and a press check?

A: A proof is a sample (PDF or printed) approved BEFORE the run. A press check is standing at the press DURING startup to approve color on actual production sheets — used for critical brand work.

Q: What does 'make-ready' mean?

A: All press setup before sellable copies: mounting plates, washing up previous inks, loading stock, and running waste sheets while adjusting color/registration.

Q: GSM vs lb vs pt — which should I use?

A: GSM. It's absolute and universal. US pounds depend on paper category (80 lb text ≠ 80 lb cover), and points measure thickness, not weight.

Q: What is variable data printing (VDP)?

A: A digital-press capability where each printed piece differs — names, addresses, codes, images — driven by a data file merged into the design. Impossible on plate-based presses.

Q: What is ganging?

A: Combining several different jobs on one press sheet (same stock/colors) so they share setup and paper cost — how online printers offer cheap business cards.

Q: Why did my printed colors look different from my screen?

A: Screens are RGB backlit; print is CMYK reflective. Without color management and CMYK-aware design, vivid screen colors fall outside the printable gamut and shift.

Q: What's the difference between Print MIS and Web-to-Print?

A: W2P is the customer-facing online store/portal (order intake). MIS is the internal engine (estimating, tickets, scheduling, invoicing). Best case: W2P feeds the MIS automatically.

Q: What are JDF and JMF in one line each?

A: JDF: an XML file describing the job's specs and process steps, exchanged between systems. JMF: the live messaging channel for machine status and counts.

Q: Where do print shops lose the most money?

A: Reprints from uncaught file errors, stale estimating rates, forgotten finishing operations in quotes, and idle machines from poor scheduling — all fixable with workflow discipline plus software.

Chapter 16 — Final Revision: The One-Stop Cheat Sheet

The master flow

Estimate → Order/Job Ticket → Preflight → Proof/Approval → Imposition → RIP/Plates → Press → Finishing → QC/Dispatch → Invoice → Estimate-vs-Actual.

Numbers to remember

Number	Meaning
300 DPI	Minimum image resolution at final print size
3 mm (1/8")	Standard bleed beyond trim; also safe-zone inset for text
4/4, 4/0, 1/0	Colors front/back shorthand (full color both sides, etc.)
~500–2,000 copies	Typical digital ↔ offset break-even zone
2–10%	Typical paper waste/overs allowance for setup
90–120 GSM	Inner pages / letterheads; 300–400 GSM = business cards & board
60/40/40/100	A common rich-black CMYK recipe for large solids

Term flash-glossary

Term	One-line meaning
Substrate	The material being printed on
Preflight	Automated file check before production
Bleed / Trim / Safe zone	Ink past the cut / the cut line / keep-content-inside line
Imposition / N-up / Ganging	Arranging pages on the press sheet / copies per sheet / mixing jobs on a sheet
Signature	A large sheet folded into a group of pages
RIP	Software converting PDFs into press dot patterns
CTP / Plate / Blanket	Computer-to-plate machine / image carrier / rubber transfer cylinder (offset)
Make-ready	Press setup time and waste before good copies
Click charge	Per-page cost on digital presses
VDP	Variable data printing — every copy different
Anilox	Engraved ink-metering roller in flexo
Saddle stitch / Perfect bind / Wire-O	Stapled booklet / glued square spine / metal ring binding
Score / Crease	Pressed hinge line so thick stock folds without cracking
Spot UV / Foil / Emboss	Gloss highlights / metallic stamping / raised surface
Dieline	Template of all cuts and creases for a die-cut piece
Job ticket	The instruction sheet traveling with a job

Term	One-line meaning
MIS / W2P	Shop management system / online ordering storefront
JDF / JMF / PDF-X	Job-spec XML / machine messaging / print-ready PDF standards
Estimate vs Actual	Comparing quoted cost to real recorded cost per job

The software builder's mental model

- A print shop = a job-shop factory: every order is custom manufacturing.
- Data skeleton: **Customer** → **Quote** → **Job** → **Operations[]** → **Events (scans)** → **Costs** → **Invoice**.
- Estimating = configurable rate cards (machine speeds, hourly rates, click rates, paper prices, waste %) + route comparison.
- Scheduling = jobs competing for machines; batching similar work is the big win.
- Automation = removing manual handoffs: intake → preflight → approval → imposition → press queue → tracking → invoice.
- Integrate with, don't fight: prepress workflows (RIP), accounting, carriers — via APIs and JDF/JMF where hardware demands it.

— End of guide. You now know more about how a print shop works than many people who walk its floor. Good luck building!

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